Parametric Constructions, Quadratic Norm Identities, and Birational Geometry of the Diophantine Equation $a^4 + b^4 + c^4 + d^4 + e^4 = (a + b + c + d + e)^4$ over $\mathbb{Z} \setminus \{0\}$

Comprehensive Mathematical Research Paper

Complete Synthesis of Sections 1–17, Algebraic Proofs, and Solution Classification

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Abstract

This paper presents a complete algebraic, number-theoretic, and geometric investigation of primitive non-zero integer solutions to the fourth-power Diophantine equation

$$a^4 + b^4 + c^4 + d^4 + e^4 = (a + b + c + d + e)^4 \quad \text{over } \mathbb{Z} \setminus \{0\}.$$

Although the projective quartic threefold $V \subset \mathbb{P}^4$ defined by this equation bounds a strictly convex compact body in the affine chart $a + b + c + d + e = 1$—and consequently contains no real straight lines—it admits a rich hierarchy of quadratic norm-form decompositions, bitangent hyperplane sections, conic bundles, and osculating rational maps. Starting from the empirical census of 42 primitive solutions with $0 < a + b + c + d + e < 1000$ (36 solutions with $5 \mid s$ and 6 solutions with $5 \nmid s$), we establish 17 interconnected structural findings and explicit parametric constructions with full algebraic proofs. We prove that $V(\mathbb{C})$ contains conjugate pairs of straight lines over $\mathbb{Q}(\sqrt{-3})$ whose common tangent hyperplanes H_{e_0} are defined over $\mathbb{Q}$ and slice V in conic bundles. We derive explicit quadratic polynomial parametrizations over genus-1 curves for all four rational bitangent sectors present in the census ($e_0 = -1, 2, \frac{3}{5}, \frac{1}{3}$), establishing in particular that both the $5 \mid s$ and $5 \nmid s$ sectors contain infinitely many primitive solutions in $\mathbb{Z} \setminus \{0\}$. Furthermore, we establish the Three-Squares Identity, the Four-Squares Surface S_0 ($2a + b + c = 0$), the 3-parameter conic-bundle family in (e_0, m, t), the 6-point and 5-point osculating conic maps around arbitrary base solutions, the rational elimination of d, the Centered Double-Depressed-Cubic Canonical Form, the Paired-Solution Surface ($u = 2e$), and the Opposite-Pair ($e = -d$) and Equal-Pair ($d = e$) surfaces.

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Empirical Census of Primitive Solutions ($0 < a+b+c+d+e < 1000$)

Before developing the general parametric theory in Sections 1–17, we record the complete empirical table of 42 primitive integer solutions to $a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4$ over $\mathbb{Z}\setminus\{0\}$ with $0<s=a+b+c+d+e<1000$, partitioned according to divisibility of s by 5.

Table I: Primitive Solutions with $a+b+c+d+e$ Divisible by 5 (36 Solutions)

Row	s	a	b	c	d	e	Row	s	a	b	c	d	e
1	5	−3	4	4	2	−2	19	585	97	514	432	−332	−126
2	15	13	12	−6	−2	−2	20	625	497	488	−432	76	−4
3	15	−9	14	8	6	−4	21	625	619	−268	156	62	56
4	135	111	−102	92	26	8	22	645	317	582	−468	116	98
5	135	111	114	−58	−16	−16	23	645	637	−276	226	64	−6
6	175	173	−72	62	26	−14	24	685	37	664	382	−252	−146
7	205	−121	182	146	−18	16	25	705	−447	644	432	128	−52
8	225	57	186	184	−104	−98	26	705	667	−416	372	54	28
9	235	73	216	−156	126	−24	27	715	681	392	−388	28	2
10	295	143	−262	216	124	74	28	755	289	748	−262	−108	88
11	295	277	−198	104	74	38	29	775	−657	604	418	322	88
12	315	−253	264	168	98	38	30	835	−117	−674	628	502	496
13	325	111	294	−242	96	66	31	835	587	668	−572	474	−322
14	355	−271	298	224	96	8	32	885	429	−808	554	488	222
15	425	−63	344	312	−306	138	33	905	−583	834	514	202	−62
16	435	123	368	−354	194	104	34	945	811	724	−546	−176	132
17	515	203	458	−394	144	104	35	965	−697	784	706	264	−92
18	515	−407	352	346	338	−114	36	985	291	936	576	−426	−392

Table II: Primitive Solutions with $a+b+c+d+e$ Not Divisible by 5 (6 Solutions)

Row	s	a	b	c	d	e	$s-a$
1 (37)	363	−262	315	230	50	30	$625=5^4$
2 (38)	413	−212	55	390	250	−70	$625=5^4$
3 (39)	507	−118	395	450	−150	−70	$625=5^4$
4 (40)	551	−74	335	490	−380	180	$625=5^4$
5 (41)	967	342	−465	900	600	−410	$625=5^4$
6 (42)	991	−259	930	−590	500	410	$1250=2\cdot5^4$

1 Explicit Elliptic-Conic Parametrization ($e_0=-1$ Sector)

We begin by presenting a complete, self-contained quadratic polynomial parametrization over a genus-1 curve that produces infinitely many primitive non-zero integer solutions to $a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4$.

Theorem 1.1 (Explicit Parametrization in the $e_0=-1$ Bitangent Sector)**.** *Define the five polynomials (a,b,c,d,e) in the rational variables (u,v) by*

$$\begin{aligned} a(u,v) &= 8u^2-76u-432,\\ b(u,v) &= -5u^2+38u+270+v,\\ c(u,v) &= -5u^2+38u+270-v,\\ d(u,v) &= 4u^2-216,\\ e(u,v) &= 8u^2+76u-432, \end{aligned} \tag{1}$$

whose sum is

$$s(u,v)=a+b+c+d+e=10u^2+76u-540. \tag{2}$$

*Then the fourth-power relation $a^4+b^4+c^4+d^4+e^4=(a+b+c+d+e)^4$ holds **identically** whenever $(u,v)\in\mathbb{Q}^2$ lies on the elliptic curve*

$$\mathcal{E}:\quad v^2=-151u^4+1140u^3+3768u^2-61560u+2916. \tag{3}$$

Furthermore, the rational point $P_0=(0,54)\in\mathcal{E}(\mathbb{Q})$ has infinite order, so the chord-and-tangent group law on $\mathcal{E}(\mathbb{Q})$ generates infinitely many primitive integer solutions in $\mathbb{Z}\setminus\{0\}$.

Proof. Set $x=\frac{5u^2-38u-270}{2(u^2-54)}$ and $y=\frac{v}{2(u^2-54)}$. Relative to the homogeneous factor $z=1$, the expressions in (1) are obtained by clearing the common denominator $-2(u^2-54)$ from the linear system

$$a=2x-1,\quad b=-x+y,\quad c=-x-y,\quad d=2,\quad e=-2x+9,\quad s=-2x+10.$$

Direct expansion of $\frac{1}{2}\big[s^4-(a^4+b^4+c^4+d^4+e^4)\big]$ in (x,y) yields

$$\begin{aligned}\frac{s^4-(a^4+b^4+c^4+d^4+e^4)}{2}&=\frac{(-2x+10)^4-(2x-1)^4-2^4-(-2x+9)^4}{2}-\frac{(-x+y)^4+(-x-y)^4}{2}\\&=\big(207x^2-1080x+1711\big)-\big(y^4+6x^2y^2-9x^4\big)\\&=\big(216x^2-1080x+1711\big)-\big(3x^2+y^2\big)^2.\end{aligned}$$

Substituting $x=\frac{5u^2-38u-270}{2(u^2-54)}$ gives

$$216x^2-1080x+1711=\left(\frac{19(u^2+54)}{u^2-54}\right)^2,$$

while $3x^2+y^2=\frac{3(5u^2-38u-270)^2+4v^2}{4(u^2-54)^2}$. Consequently, $\big(3x^2+y^2\big)^2=216x^2-1080x+1711$ holds if and only if

$$3(5u^2-38u-270)^2+4v^2=4(u^2-54)\cdot\big[-19(u^2+54)\big],$$

which simplifies upon dividing by 4 to $v^2=-151u^4+1140u^3+3768u^2-61560u+2916$. □

Explicit Generation of Primitive Solutions from Theorem 1.1

1. **Base Point ($u=0, v=54$)**: Substituting $(u,v)=(0,54)$ into (1) yields $(-432,324,216,-216,-432)$. Dividing by -108 gives Row 1 of Table I (up to permutation):

$$(a,b,c,d,e)=(4,\,-3,\,-2,\,2,\,4),\qquad s=5.$$

2. **First Osculating Tangent Step ($P_1=-2P_0$)**: Expanding $\sqrt{-151u^4+1140u^3+3768u^2-61560u+54^2}$ to second order at $u=0$ defines the osculating parabola

$$v(u)=54-570u-\frac{26761}{36}u^2.$$

Then $v(u)^2-(-151u^4+1140u^3+3768u^2-61560u+2916)=\frac{722}{81}u^3(247979u+95040)=0$. The non-zero root is

$$u=-\frac{95040}{247979},\qquad v=\frac{10035361202634}{61493584441}.$$

Substituting (u,v) into (1) and clearing denominators and the greatest common divisor yields the primitive solution:

$$\begin{aligned}&a=12037916876,\quad b=-2709042377,\quad c=-12556575238,\\&d=6455401478,\quad e=13783689036,\quad s=17011389775.\end{aligned}$$

2 Strict Convexity Obstruction, Norm-Form Identities, and Bitangent Geometry

In this section we prove why no purely linear (rational line) parametrization exists on $V(\mathbb{R})$, establish the fundamental Three-Squares and Four-Squares Identities, and derive the bitangent hyperplane family H_{e_0}.

2.1 Strict Convexity of the Real Quartic Threefold

Proposition 2.1 (Strict Convexity and Absence of Real Lines)**.** *In the affine chart $s=a+b+c+d+e=1$, the real locus $V(\mathbb{R})$ is defined by $g(a,b,c,d)=a^4+b^4+c^4+d^4+(1-a-b-c-d)^4=1$. Its Hessian matrix is strictly positive definite at every point of $V(\mathbb{R})$ having at most one zero coordinate. Consequently:*

1. *$V(\mathbb{R})$ bounds a strictly convex compact body in $\mathbb{R}^4$ and contains **no real straight lines**.*
2. *For any real point $P\in V(\mathbb{R})$ with non-zero coordinates, the real tangent hyperplane $T_PV(\mathbb{R})$ meets $V(\mathbb{R})$ solely at P.*
3. *Any real 2-plane $\Pi\subset\mathbb{P}^4(\mathbb{R})$ intersecting $V(\mathbb{R})$ in more than one real point can only be tangent to $V(\mathbb{C})$ at non-real complex conjugate pairs of points $P,\bar{P}\in V(\mathbb{C})$.*

Proof. Computing the second partial derivatives of $g(x_1,x_2,x_3,x_4)=\sum_{i=1}^4 x_i^4+x_5^4$ where $x_5=1-\sum_{i=1}^4 x_i$ gives $\frac{\partial^2 g}{\partial x_i\partial x_j}=12x_i^2\delta_{ij}+12x_5^2$, so for any non-zero vector $v=(v_1,v_2,v_3,v_4)^T\in\mathbb{R}^4$,

$$v^T(D^2g)v=12\left(\sum_{i=1}^4 x_i^2v_i^2+x_5^2\left(\sum_{i=1}^4 v_i\right)^2\right)>0$$

whenever at most one of $x_1,\dots,x_5$ vanishes. Equivalently, in homogeneous coordinates, at any real point $P=(x_1,\dots,x_5)\in V(\mathbb{R})$, a tangent direction v satisfies $\sum_{i=1}^5 x_i^3v_i=s^3v_s$ (with $v_s=\sum_{i=1}^5 v_i$). By the Cauchy–Schwarz inequality,

$$(s^3v_s)^2=\left(\sum_{i=1}^5(x_i^2)(x_iv_i)\right)^2\le\left(\sum_{i=1}^5 x_i^4\right)\left(\sum_{i=1}^5 x_i^2v_i^2\right)=s^4\sum_{i=1}^5 x_i^2v_i^2,$$

so $\sum_{i=1}^5 x_i^2v_i^2-s^2v_s^2\ge 0$, with equality if and only if v is proportional to P (which is $v=0$ in $T_P\mathbb{P}^4$). Thus the second fundamental form is definite, forbidding real inflectional tangents and real lines. Since non-real singularities of a real plane quartic $C=V\cap\Pi$ occur in conjugate pairs, the total delta-invariant $\delta=\sum\delta_P$ of any real plane section C containing a real oval is necessarily even ($\delta\in\{0,2,4\}$), giving genus $g=3-\delta\in\{3,1\}$ or a pair of conics ($\delta=4$). □

2.2 The Three-Squares and Four-Squares Identities

Theorem 2.2 (Three-Squares Identity and Four-Squares Surface S_0)**.** *Write $b=-x+y$, $c=-x-y$, $d=-z+w$, and $e=-z-w$ (so $x=-\frac{b+c}{2}$, $y=\frac{b-c}{2}$, $z=-\frac{d+e}{2}$, $w=\frac{d-e}{2}$, and $s=a-2x-2z$). Then the fourth-power equation $a^4+b^4+c^4+d^4+e^4=s^4$ is **identically equivalent** to the Three-Squares Identity:*

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2-\left(4(x^2+xz+z^2)-2a(x+z)+a^2\right)^2=-a\left(a^3-8axz+16xz(x+z)\right). \tag{4}$$

In particular, on the hyperplane $2a+b+c=0$ (i.e., $a=x$), the cubic remainder factors as $-x^2(x+4z)^2$, reducing the equation to the ***Four-Squares Surface*** *S_0:*

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2+x^2(x+4z)^2=\left(3x^2+2xz+4z^2\right)^2, \tag{5}$$

or equivalently, in factored norm form:

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2=4\left(x^2-xz+2z^2\right)\left(2x^2+3xz+2z^2\right). \tag{6}$$

Proof. Since $(-x+y)^4+(-x-y)^4=2x^4+12x^2y^2+2y^4=2(y^2+3x^2)^2-16x^4$ and similarly $d^4+e^4=2(w^2+3z^2)^2-16z^4$, the equation $a^4+b^4+c^4+d^4+e^4-(a-2x-2z)^4=0$ divides by 2 to give

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2=\frac{(a-2x-2z)^4-a^4}{2}+8x^4+8z^4.$$

Expanding $\left(4(x^2+xz+z^2)-2a(x+z)+a^2\right)^2$ and subtracting from the right-hand side yields

$$\begin{aligned}&\left[-4a^3(x+z)+12a^2(x+z)^2-16a(x+z)^3+16(x^2+xz+z^2)^2\right]\\&\quad-\left[4(x^2+xz+z^2)-2a(x+z)+a^2\right]^2\\&\quad=-a^4+8a^2xz-16axz(x+z)=-a\left(a^3-8axz+16xz(x+z)\right),\end{aligned}$$

proving (4). Setting $a=x$ gives $a^3-8axz+16xz(x+z)=x(x^2+8xz+16z^2)=x(x+4z)^2$ and $4(x^2+xz+z^2)-2x(x+z)+x^2=3x^2+2xz+4z^2$, establishing (5) and (6). □

Remark 2.3 (Symmetric Newton Identity Form). In terms of $p=d+e$, $q=de$, $\sigma_1=a+b+c$, $\sigma_2=ab+bc+ca$, and $\sigma_3=abc$, Newton's identities also yield the equivalent decomposition

$$\begin{aligned}\frac{(a+b+c+d+e)^4-(a^4+b^4+c^4+d^4+e^4)}{2}&=\left(p^2+p\sigma_1+\sigma_1^2\right)^2-\left(d^2+de+e^2\right)^2\\&\quad-\left(\sigma_1^2-\sigma_2\right)^2-2\sigma_1\sigma_3=0,\end{aligned}$$

and for two pairs $p_1=a+b$, $p_2=c+d$:

$$\left(a^2+ab+b^2\right)^2+\left(c^2+cd+d^2\right)^2-\left(p_1^2+p_1p_2+p_2^2+e(p_1+p_2)\right)^2=2e(e+p_1)(e+p_2)(p_1+p_2).$$

2.3 Bitangent Hyperplanes via Complex Conjugate Lines on $V(\mathbb{C})$

Theorem 2.4 (Rational Bitangent Hyperplane Family H_{e_0})**.** *Let $\omega=\frac{-1+i\sqrt{3}}{2}$ be a primitive cube root of unity. For any $e_0\in\mathbb{Q}$, the complex conjugate lines*

$$L_{e_0}=\left\{(t,\,\omega t,\,\omega^2 t,\,0,\,e_0t+w)\right\},\qquad \bar{L}_{e_0}=\left\{(t,\,\omega^2 t,\,\omega t,\,0,\,e_0t+w)\right\}$$

lie on $V(\mathbb{C})$ when $w=0$. At the conjugate points $P=(1,\omega,\omega^2,0,e_0)$ and $\bar{P}=(1,\omega^2,\omega,0,e_0)$, the tangent hyperplanes $T_PV(\mathbb{C})$ and $T_{\bar{P}}V(\mathbb{C})$ ***coincide and are defined over*** *$\mathbb{Q}$:*

$$H_{e_0}:\quad (e_0^3-1)(a+b+c)+e_0^3d=0\iff\frac{a+b+c}{a+b+c+d}=e_0^3. \tag{7}$$

Parametrizing H_{e_0} in coordinates $(x:y:z:w)\in\mathbb{P}^3$ by

$$a=2x+e_0^3z,\quad b=-x+y,\quad c=-x-y,\quad d=(1-e_0^3)z,\quad e=2e_0x+w\quad(s=z+e) \tag{8}$$

cancels both the x^4 and x^3 terms of $s^4-(a^4+b^4+c^4+d^4+e^4)=0$, reducing the quartic surface $S_{e_0}=V\cap H_{e_0}$ to a quadratic conic bundle in x:

$$\left(3x^2+y^2\right)^2=z\left[12e_0^2\left(2w+(1-e_0^4)z\right)x^2+4e_0\left(3w^2+3wz+(1-e_0^8)z^2\right)x+C_{e_0}(w,z)\right], \tag{9}$$

where $C_{e_0}(w,z)=2w^3+3w^2z+2wz^2-e_0^3(e_0^9-2e_0^6+3e_0^3-2)z^3$.

Proof. At $P=(1,\omega,\omega^2,0,e_0)$, we have $s=1+\omega+\omega^2+0+e_0=e_0$ and $a^3=b^3=c^3=1$, $d^3=0$, $e^3=e_0^3=s^3$. Since $\frac{\partial F}{\partial x_k}=4(s^3-x_k^3)$, the gradient at both P and $\bar{P}$ is

$$\nabla F(P)=\nabla F(\bar{P})=4\left(e_0^3-1,\, e_0^3-1,\, e_0^3-1,\, e_0^3,\, 0\right),$$

yielding (7). Under (8), $b^4+c^4=2(3x^2+y^2)^2-16x^4$, while $a^4=(2x+e_0^3z)^4=16x^4+32e_0^3x^3z+\dots$ cancels $-16x^4$, and $s^4-e^4=(z+2e_0x+w)^4-(2e_0x+w)^4=4z(2e_0x+w)^3+\dots$ has leading x^3 term $32e_0^3x^3z$, which cancels the $32e_0^3x^3z$ term from a^4, leaving (9). □

3 Number-Theoretic Structure of the Solutions Table (mod 5 Dichotomy)

Theorem 3.1 (5-Adic Divisibility Law)**.** *Let $(a,b,c,d,e)\in(\mathbb{Z}\setminus\{0\})^5$ be a primitive solution to $a^4+b^4+c^4+d^4+e^4=s^4$ with $s=a+b+c+d+e$.*

1. ***Case 1 ($5\mid s$):*** *None of a,b,c,d,e is divisible by 5: $a^4\equiv b^4\equiv c^4\equiv d^4\equiv e^4\equiv 1 \pmod 5$.*
2. ***Case 2 ($5\nmid s$):*** *Exactly four of the variables (say b,c,d,e) are divisible by 5, while the remaining variable satisfies $a\equiv s\not\equiv 0 \pmod 5$ and the strict 5^4-divisibility law:*

$$625=5^4\,\Big|\,(s-a)=b+c+d+e. \tag{10}$$

Proof. By Fermat's Little Theorem, $x^4\equiv 0 \pmod 5$ if $5\mid x$ and $x^4\equiv 1 \pmod 5$ if $5\nmid x$. Modulo 5, the sum of five fourth powers is congruent to the number of terms not divisible by 5. If $5\mid s$, then $s^4\equiv 0 \pmod 5$, so either zero or all five of (a,b,c,d,e) are divisible by 5; by primitivity, none is divisible by 5. If $5\nmid s$, then $s^4\equiv 1 \pmod 5$, so exactly one variable (say a) is not divisible by 5, while $5\mid b,c,d,e$. Then $s=a+b+c+d+e\equiv a \pmod 5$, and

$$(s-a)(s+a)(s^2+a^2)=s^4-a^4=b^4+c^4+d^4+e^4\equiv 0 \pmod{5^4}.$$

Since $s\equiv a\not\equiv 0 \pmod 5$, we have $s+a\equiv 2s\not\equiv 0 \pmod 5$ and $s^2+a^2\equiv 2s^2\not\equiv 0 \pmod 5$. Thus $\gcd\big((s+a)(s^2+a^2),\,5\big)=1$, forcing $5^4=625\mid(s-a)$. This explains why $s-a=625$ in Rows 1–5 of Table II and $s-a=1250=2\cdot 625$ in Row 6 of Table II. □

4 General 3-Parameter Conic-Bundle Parametrization in (e_0,m,t)

We now extend Theorem 1.1 to a 3-parameter family on H_{e_0} that requires no prior point to parameterize the conic bundle.

Theorem 4.1 (General 3-Parameter Conic-Bundle Family)**.** *For any rational parameters $e_0\in\mathbb{Q}\setminus\{0,\pm1\}$ and $m\in\mathbb{Q}$, set $w=\frac{e_0^4-1+3m^2}{2}z$ in Theorem 2.4. Then the coefficient of x^2z^2 on the right-hand side of (9) becomes identically a perfect square, $A_0=(6e_0m)^2$, and the quartic surface S_{e_0} reduces to*

$$\left(3x^2+y^2\right)^2=z^2\left[(6e_0m)^2x^2+B_0(e_0,m)xz+C_0(e_0,m)z^2\right], \tag{11}$$

where

$$B_0(e_0,m)=-e_0\left(e_0^8-18e_0^4m^2-27m^4-1\right), \tag{12}$$

$$C_0(e_0,m)=\frac{27m^6+27e_0^4m^4+3(3e_0^8+1)m^2-3e_0^{12}+8e_0^9-12e_0^6+e_0^4+8e_0^3-2}{4}. \tag{13}$$

Setting $3x^2+y^2=(6e_0mx+tz)z$ solves linearly for $\frac{x}{z}=\frac{C_0-t^2}{12e_0mt-B_0}$, giving the explicit 3-parameter polynomial family:

$$\begin{aligned} a(e_0,m;t) &= 2\left(C_0-t^2\right)+e_0^3\left(12e_0mt-B_0\right),\\ b(e_0,m;t,Y) &= -\left(C_0-t^2\right)+Y,\\ c(e_0,m;t,Y) &= -\left(C_0-t^2\right)-Y,\\ d(e_0,m;t) &= (1-e_0^3)\left(12e_0mt-B_0\right),\\ e(e_0,m;t) &= 2e_0\left(C_0-t^2\right)+\frac{e_0^4-1+3m^2}{2}\left(12e_0mt-B_0\right), \end{aligned} \tag{14}$$

where (t,Y) lies on the 2-parameter family of genus-1 curves

$$\mathcal{E}_{e_0,m}:\quad Y^2=(12e_0mt-B_0)\left[6e_0m(C_0-t^2)+t(12e_0mt-B_0)\right]-3\left(C_0-t^2\right)^2. \tag{15}$$

Proof. In (9), the coefficient of x^2 (for $z=1$) is $12e_0^2(2w+1-e_0^4)$. Substituting $2w+1-e_0^4=3m^2$ yields $12e_0^2(3m^2)=(6e_0m)^2$. Expanding $(6e_0mx+tz)^2=36e_0^2m^2x^2+12e_0mtxz+t^2z^2$ and equating to $36e_0^2m^2x^2+B_0xz+C_0z^2$ cancels $36e_0^2m^2x^2$ and leaves $(12e_0mt-B_0)x=(C_0-t^2)z$. Substituting into $y^2=(6e_0mx+tz)z-3x^2$ and clearing $(12e_0mt-B_0)^2$ gives $\mathcal{E}_{e_0,m}$. □

Corollary 4.2 (Linear Degeneration at $m=0$)**.** *When $m=0$ (i.e., $w=\frac{e_0^4-1}{2}z$), the x^2 term vanishes completely ($A_0=0$), and S_{e_0} becomes linear in x:*

$$\left(3x^2+y^2\right)^2=z^3\left(B_0(e_0,0)x+C_0(e_0,0)z\right),$$

where $B_0(e_0,0)=-e_0(e_0^8-1)$ and $C_0(e_0,0)=-\frac{3e_0^{12}-8e_0^9+12e_0^6-e_0^4-8e_0^3+2}{4}$. Setting $3x^2+y^2=uz^2$ expresses x/z directly as a quadratic polynomial in u for every $e_0\in\mathbb{Q}\setminus\{0,\pm1\}$:

$$\frac{x}{z}=\frac{u^2-C_0(e_0,0)}{B_0(e_0,0)},\qquad \left(\frac{y}{z}\right)^2=u-3\left(\frac{u^2-C_0(e_0,0)}{B_0(e_0,0)}\right)^2.$$

5 Universal Bitangent-Plane Parametrization and the $e_0=2$ Sector

Theorem 5.1 (Universal Bitangent-Plane Parametrization from Any Base Point on H_{e_0})**.** *Let $P_0=(a_0,b_0,c_0,d_0,e_0^*)$ be any solution satisfying $\frac{a_0+b_0+c_0}{a_0+b_0+c_0+d_0}=e_0^3$ for some $e_0\in\mathbb{Q}$. Set*

$$z_0=a_0+b_0+c_0+d_0,\quad x_0=-\frac{b_0+c_0}{2},\quad y_0=\frac{b_0-c_0}{2},\quad w_0=e_0^*-2e_0x_0,\quad U_0=3x_0^2+y_0^2.$$

Fixing $z=z_0$ and $w=w_0$ in (9) defines a conic $U^2=\mathcal{A}x^2+\mathcal{B}x+\mathcal{C}$ passing through (x_0,U_0). Parameterizing this conic via $U-U_0=t(x-x_0)$ yields the universal quadratic polynomial parametrization:

$$\begin{aligned} a(t) &= 2\left(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B}\right)+e_0^3z_0(t^2-\mathcal{A}),\\ b(t,Y) &= -\left(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B}\right)+Y,\\ c(t,Y) &= -\left(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B}\right)-Y,\\ d(t) &= (1-e_0^3)z_0(t^2-\mathcal{A}),\\ e(t) &= 2e_0\left(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B}\right)+w_0(t^2-\mathcal{A}), \end{aligned} \tag{16}$$

where (t,Y) satisfies the genus-1 curve

$$Y^2=(t^2-\mathcal{A})\left(-U_0t^2+(2\mathcal{A}x_0+\mathcal{B})t-\mathcal{A}U_0\right)-3\left(x_0t^2-2U_0t+\mathcal{A}x_0+\mathcal{B}\right)^2. \tag{17}$$

Proof. Substituting $U = U_0 + t(x - x_0)$ into $U^2 - (\mathcal{A}x^2 + \mathcal{B}x + \mathcal{C}) = 0$ and using $U_0^2 = \mathcal{A}x_0^2 + \mathcal{B}x_0 + \mathcal{C}$ gives

$$(x - x_0)\left[(t^2 - \mathcal{A})x - (x_0t^2 - 2U_0t + \mathcal{A}x_0 + \mathcal{B})\right] = 0.$$

Solving the non-trivial factor for $x(t)$ and $U(t) = U_0 + t(x(t) - x_0)$ and clearing the denominator $t^2 - \mathcal{A}$ in (8) establishes (16). □

Example 5.2 (Worked Parametrization for $e_0 = 2$ from Row 6: $s = 175$)**.** In Row 6 of Table I, $(a_0, b_0, c_0, d_0, e_0^*) = (62, 26, -72, -14, 173)$ has $a_0 + b_0 + c_0 = 16$ and $a_0 + b_0 + c_0 + d_0 = 2$, so $\frac{16}{2} = 8 = 2^3 \implies e_0 = 2$. Here $z_0 = 2$, $x_0 = 23$, $y_0 = 49$, $w_0 = 81$, $U_0 = 3988$, and $(\mathcal{A}, \mathcal{B}, \mathcal{C}) = (12672, 306384, 2153824)$. Rescaling $t = 8u$ in Theorem 5.1 yields the explicit polynomial parametrization:

$$\begin{aligned}
a(u) &= 2\left(156u^2 - 3988u + 31029\right), \\
b(u, W) &= -\left(92u^2 - 3988u + 37365\right) + W, \\
c(u, W) &= -\left(92u^2 - 3988u + 37365\right) - W, \\
d(u) &= -56\left(u^2 - 198\right), \\
e(u) &= 4\left(173u^2 - 3988u + 21327\right),
\end{aligned}$$

where $\mathcal{E}_2 : W^2 = -89200u^4 + 3979968u^3 - 68337912u^2 + 541908504u - 1686900843$, with base point at the tangent slope $u_0 = \frac{55581}{3988}$.

6 General 6-Point Osculating-Parabola Map Around Symmetric Solutions ($b_0 = c_0$)

Theorem 6.1 (6-Point Osculating-Parabola Rational Map)**.** *Let $P_0 = (a_0, -x_0, -x_0, d_0, e_0)$ be any solution with two equal coordinates $b_0 = c_0 = -x_0$ (such as Row 1, Row 2, or Row 5 of Table I). For any direction vector $Q = (a_2, 0, 0, d_2, e_2)$ parameterized by $(a_2 : d_2 : e_2) \in \mathbb{P}^2(\mathbb{Q})$, consider the 2-plane $\Pi_{(a_2:d_2:e_2)}$ given by*

$$a = a_0x + a_2z, \quad b = -x_0x + y, \quad c = -x_0x - y, \quad d = d_0x + d_2z, \quad e = e_0x + e_2z.$$

Then the plane quartic $C = V \cap \Pi_{(a_2:d_2:e_2)}$ has the equation

$$\left(3x_0^2x^2 + y^2\right)^2 = 9x_0^4x^4 + c_3x^3z + c_2x^2z^2 + c_1xz^3 + c_0z^4, \tag{18}$$

where $c_k(a_2, d_2, e_2)$ is homogeneous of degree $4 - k$. Define

$$\beta = \frac{c_3}{6x_0^2}, \qquad \gamma = \frac{c_2 - \beta^2}{6x_0^2}, \qquad R_1 = c_1 - 2\beta\gamma, \qquad R_0 = c_0 - \gamma^2. \tag{19}$$

*Then the parabola $y^2 = z(\beta x + \gamma z)$ osculates C at $P_0 = (1 : 0 : 0)$ with **multiplicity** 6, and the remaining two intersection points are given over $\mathbb{P}^2_{(a_2:d_2:e_2)}$ by:*

$$\frac{x}{z} = -\frac{R_0(a_2, d_2, e_2)}{R_1(a_2, d_2, e_2)}, \qquad \left(\frac{y}{z}\right)^2 = \frac{\gamma(a_2, d_2, e_2)R_1(a_2, d_2, e_2) - \beta(a_2, d_2, e_2)R_0(a_2, d_2, e_2)}{R_1(a_2, d_2, e_2)}. \tag{20}$$

Proof. By definition of β, γ, R_1, R_0, we have the algebraic identity

$$9x_0^4x^4 + c_3x^3z + c_2x^2z^2 + c_1xz^3 + c_0z^4 = \left(3x_0^2x^2 + \beta xz + \gamma z^2\right)^2 + z^3(R_1x + R_0z).$$

Equating $\left(3x_0^2x^2 + y^2\right)^2$ to the right-hand side and imposing $3x_0^2x^2 + y^2 = 3x_0^2x^2 + \beta xz + \gamma z^2$ (which cancels $3x_0^2x^2$ to give $y^2 = \beta xz + \gamma z^2$) forces $z^3(R_1x + R_0z) = 0$. The six roots $z^3 = 0$ (each giving $y^2 = 0$, multiplicity 2) account for 6 intersection points at $P_0 = (1 : 0 : 0)$, while $R_1x + R_0z = 0$ yields $\frac{x}{z} = -\frac{R_0}{R_1}$ and $\left(\frac{y}{z}\right)^2 = \beta\frac{x}{z} + \gamma = \frac{\gamma R_1 - \beta R_0}{R_1}$. □

7 Rational Elimination of d and the Separation-of-Variables Surface

Theorem 7.1 (Rational Elimination of d in Terms of Free Parameters (a,x,e))**.** *In the Three-Squares Identity (4), setting the second and third squares equal via $d^2+de+e^2=4(x^2+xz+z^2)-2a(x+z)+a^2$ (with $z=-\frac{d+e}{2}$) cancels the d^2 term identically and solves **linearly and rationally for** d in terms of (a,x,e):*

$$d=-\frac{a^2+ae-2ax-2ex+4x^2}{a+e-2x},\qquad b=-x+y,\qquad c=-x-y. \tag{21}$$

Under (21), setting $a=-m^2x$ reduces the 5-variable fourth-power equation to a binary quartic in (e,x) whose leading e^4 coefficient is $(2mx^2)^2$ (a perfect square for every $m\in\mathbb{Q}$):

$$\left[\left(y^2+3x^2\right)\left(e-(m^2+2)x\right)\right]^2=\left(2mxe^2-m(m^2+2)x^2e-\frac{m(m^2+5)(m^4+4)}{4}x^3\right)^2+R_1(m)x^5e+R_0(m)x^6, \tag{22}$$

where

$$R_1(m)=\frac{3}{2}m^2(m^2+1)(m^2+2)(m^4+4), \tag{23}$$

$$R_0(m)=-\frac{1}{16}m^2(m^2+1)(m^4+4)\left(m^6+25m^4+68m^2+100\right), \tag{24}$$

yielding the explicit rational specialization $\frac{e}{x}=-\frac{R_0(m)}{R_1(m)}=\frac{m^6+25m^4+68m^2+100}{24(m^2+2)}$.

Proof. Substituting $z=-\frac{d+e}{2}$ into $4(x^2+xz+z^2)-2a(x+z)+a^2-(d^2+de+e^2)=0$ cancels d^2+de+e^2 against $4z^2-de=(d+e)^2-de$ and leaves $(a-2x+e)d+(a^2+ae-2ax-2ex+4x^2)=0$, giving (21). Substituting $z=\frac{e^2-(a^2-2ax+4x^2)}{2(a+e-2x)}$ into $-a(a^3-8axz+16xz(x+z))$ with $a=-m^2x$ yields a degree-4 polynomial in e with leading term $-a(4xe^4)=4m^2x^2e^4=(2mxe^2)^2$. Completing the square in e gives the stated $R_1(m)$ and $R_0(m)$, where the common factor $m^2(m^2+1)(m^4+4)$ arises because the quadratic in z has zero discriminant when $a=x$ ($m^2=-1$) or $a^2+4x^2=0$ ($m^4=-4$). □

Theorem 7.2 (General Separation-of-Variables Surface)**.** *For any solution (a,b,c,d,e), set $b=-x+y$, $c=-x-y$, and $d=2x+p$ (so $p=b+c+d$ and $s=a+p+e$). Then the $16x^4$ terms cancel between b^4+c^4 and d^4, and the variables a and x separate completely:*

$$2\left(y^2+3x^2\right)^2=\left[(a+p+e)^4-a^4-p^4-e^4\right]-\left[(2x+p)^4-(2x)^4-p^4\right]. \tag{25}$$

In particular, on the hyperplane $p=2e$ (i.e., $s-a=3e$, satisfied by Rows 1, 2, 4, 16, and 17 of Table I), (25) simplifies to the separated cubic surface:

$$\left(y^2+3x^2\right)^2=e\left(6a^3+27ea^2+54e^2a+32e^3-32x^3-48ex^2-32e^2x\right). \tag{26}$$

Proof. Since $b^4+c^4+d^4=2(y^2+3x^2)^2-16x^4+(2x+p)^4=2(y^2+3x^2)^2+\left[(2x+p)^4-(2x)^4-p^4\right]+p^4$, subtracting from $s^4-a^4-e^4=(a+p+e)^4-a^4-e^4$ immediately yields (25). □

8 Universal Tangent-Slope Shift and Closed-Form Rational Recurrence

Theorem 8.1 (Universal Tangent-Slope Shift and Closed-Form Recurrence Step)**.** *In Theorem 5.1, when the conic $U^2=\mathcal{A}x^2+\mathcal{B}x+\mathcal{C}$ is parameterized by $U-U_0=t(x-x_0)$, the base solution (x_0,y_0,U_0) with $U(t_0)=+U_0>0$ is located at the tangent slope*

$$t_0=\frac{2\mathcal{A}x_0+\mathcal{B}}{2U_0}. \tag{27}$$

Shifting the parameter via $t=t_0+\kappa u$ (for any scaling constant $\kappa\in\mathbb{Q}^\times$) places the base solution at $u=0$ on a genus-1 quartic of the form

$$\mathcal{E}: \quad Y^2=c_4u^4+c_3u^3+c_2u^2+c_1u+E_0^2 \qquad (E_0>0). \tag{28}$$

The osculating parabola $Y(u)=E_0+m_1u+m_2u^2$ at $(0,E_0)$ with

$$m_1=\frac{c_1}{2E_0}, \qquad m_2=\frac{4E_0^2c_2-c_1^2}{8E_0^3}$$

cancels the u^0,u^1,u^2 coefficients of $Y(u)^2-\mathcal{E}(u)=0$ identically and yields the closed-form next rational point $(u_{new},Y_{new})=-2(0,E_0)$:

$$u_{new}=\frac{8E_0^2\left(8E_0^4c_3-4E_0^2c_1c_2+c_1^3\right)}{-64E_0^6c_4+16E_0^4c_2^2-8E_0^2c_1^2c_2+c_1^4}, \qquad Y_{new}=E_0+m_1u_{new}+m_2u_{new}^2. \tag{29}$$

Proof. Differentiating $U^2=\mathcal{A}x^2+\mathcal{B}x+\mathcal{C}$ at (x_0,U_0) gives $2U_0\frac{dU}{dx}=2\mathcal{A}x_0+\mathcal{B}$, so the tangent line at (x_0,U_0) has slope $t_0=\frac{2\mathcal{A}x_0+\mathcal{B}}{2U_0}$, where the second intersection point on the conic coincides with $(x_0,+U_0)$ and $y^2(t_0)=U_0-3x_0^2=+y_0^2>0$. Expanding $(E_0+m_1u+m_2u^2)^2-(c_4u^4+c_3u^3+c_2u^2+c_1u+E_0^2)$ with the given m_1,m_2 leaves $u^3\left[(m_2^2-c_4)u+(2m_1m_2-c_3)\right]=0$, whose non-zero root simplifies directly to (29). □

9 Explicit Parametrization for the $5\nmid(a+b+c+d+e)$ Sector $\left(e_0=\frac{3}{5}\right)$

We now prove that the second table ($5\nmid(a+b+c+d+e)$) also contains infinitely many primitive solutions by constructing an explicit polynomial parametrization from Row 4 of Table II.

Theorem 9.1 (Explicit Polynomial Parametrization for the $5\nmid s$ Sector)**.** *Define (a,b,c,d,e) in terms of (t,W) by*

$$\begin{aligned}
a(t)&=180t^2-51730t+5927700,\\
b(t,W)&=\frac{-5\left(9t^2-10346t+1434858\right)+5W}{2},\\
c(t,W)&=\frac{-5\left(9t^2-10346t+1434858\right)-5W}{2},\\
d(t)&=490\left(t^2-9234\right),\\
e(t)&=-74t^2-31038t+5237208,
\end{aligned} \tag{30}$$

*with sum $s(t)=551t^2-31038t+2327028$. Then $a^4+b^4+c^4+d^4+e^4=s^4$ holds **identically** whenever (t,W) lies on the genus-1 curve*

$$\mathcal{E}_{3/5}: \quad W^2=-5589t^4+1473452t^3-73815088t^2+1810849212t-201742694346. \tag{31}$$

Shifting $t=\frac{108426}{739}+u$ places Row 4 of Table II, $(180,335,-380,490,-74)$ with $s=551$, at $u=0$, and the recurrence step (29) yields

$$u_{new}=\frac{34296822315976713695781939 36}{61669152348095808019404623},$$

producing a 44-digit primitive solution with $s\equiv 2\not\equiv 0 \pmod 5$. Iterating the group law on $\mathcal{E}_{3/5}(\mathbb{Q})$ generates infinitely many primitive solutions with $5\nmid(a+b+c+d+e)$.

Proof. In Row 4 of Table II, $(a_0,b_0,c_0,d_0,e_0^*)=(180,335,-380,490,-74)$ with $s=551$ satisfies

$$\frac{a_0+b_0+c_0}{a_0+b_0+c_0+d_0}=\frac{180+335-380}{180+335-380+490}=\frac{135}{625}=\frac{27}{125}=\left(\frac{3}{5}\right)^3 \implies e_0=\frac{3}{5}.$$

Setting $z=625$, $x=\frac{5}{2}X$, and $y=\frac{5}{2}Y$ on $H_{3/5}$ gives $a=5X+135$, $b=\frac{-5X+5Y}{2}$, $c=\frac{-5X-5Y}{2}$, $d=490$, $e=3X-101$, and reduces (9) to

$$\left(3X^2+Y^2\right)^2=147744X^2+21628032X+221539312,$$

with base point $(X_0,Y_0)=(9,143)$ and $U_0=3(9)^2+143^2=20692$. Parameterizing the conic via $3X^2+Y^2=20692+4t(X-9)$ gives $X(t)=\frac{9t^2-10346t+1434858}{t^2-9234}$, and clearing t^2-9234 yields (30) and $\mathcal{E}_{3/5}$. □

10 General Non-Even Osculating Conic Around an Arbitrary Solution ($b_0\neq c_0$)

We can drop the symmetry requirement $b_0=c_0$ of Theorem 6.1 and construct a 5-point osculating conic around an arbitrary rational solution $P_0\in V(\mathbb{Q})$.

Theorem 10.1 (Triangular Linear System for the Non-Even Osculating Conic)**.** *Let $P_0=(a_0,b_0,c_0,d_0,e_0)\in V(\mathbb{Q})$ be any solution with $b_0\neq c_0$, normalized by scaling so that $b_0+c_0=-1$, and set $y_0=\frac{b_0-c_0}{2}\neq 0$. On the 2-plane $\Pi_{(a_2:d_2:e_2)}$ given by*

$$a=a_0x+a_2z,\quad b=-\frac{1}{2}x+y,\quad c=-\frac{1}{2}x-y,\quad d=d_0x+d_2z,\quad e=e_0x+e_2z,$$

the plane quartic $C=V\cap\Pi_{(a_2:d_2:e_2)}$ is

$$\left(\frac{3}{4}x^2+y^2\right)^2=c_4x^4+c_3x^3z+c_2x^2z^2+c_1xz^3+c_0z^4,\qquad c_4=\left(\frac{3}{4}+y_0^2\right)^2,$$

containing both $P_0=(1:y_0:0)$ and $P_2=(1:-y_0:0)$ on $z=0$. Consider the conic in $\Pi_{(a_2:d_2:e_2)}$ passing through P_0 (at $t=\infty$) and P_2 (at $t=0$):

$$x(t)=t^2+x_1t+x_0,\qquad y(t)=y_0t^2+y_1t-y_0x_0,\qquad z(t)=t. \tag{32}$$

*The conditions for the coefficients of t^7,t^6,t^5 on C to vanish (so the conic osculates C to order 4 at P_0 while also passing through P_2) decouple into a **triangular linear system** in (y_1,x_0,x_1):*

$$y_1=y_0x_1+\frac{c_3}{y_0(4y_0^2+3)}, \tag{33}$$

$$x_0=\frac{3c_3^2(4y_0^2+1)-2c_2y_0^2(4y_0^2+3)^2}{4y_0^4(4y_0^2+3)^3}, \tag{34}$$

$$x_1=\frac{162c_1y_0^4(4y_0^2+3)^3-54c_2c_3y_0^2(4y_0^2+3)(8y_0^2+1)+c_3^3(112y_0^4+48y_0^2+9)}{27y_0^2(4y_0^2+3)^2\left[2c_2y_0^2(4y_0^2+3)-c_3^2(8y_0^2+1)\right]}. \tag{35}$$

With these explicit rational formulas, 5 of the 8 intersection points of the conic with C lie at P_0 and P_2, leaving a residual cubic $d_4t^3+d_3t^2+d_2t+d_1=0$ with $d_1=-2y_0(4y_0^2+3)x_0^3y_1$.

Proof. Substituting (32) into $(\frac{3}{4}x^2+y^2)^2-(c_4x^4+c_3x^3z+c_2x^2z^2+c_1xz^3+c_0z^4)$ yields a polynomial in t whose t^8 and t^0 coefficients vanish identically. Its t^7 coefficient is $-c_3-y_0^2(4y_0^2+3)x_1+y_0(4y_0^2+3)y_1=0$, giving (33). Substituting (33) into the t^6 coefficient cancels all x_1^2 and x_1 terms identically, leaving a linear equation in x_0 whose unique solution is (34). Finally, substituting both (33) and (34) into the t^5 coefficient cancels all x_1^3 and x_1^2 terms identically, leaving a linear equation in x_1 with solution (35). □

11 Depressed-Cubic Reduction on the Separated Surface for Arbitrary $(e,p)\in\mathbb{Q}^2$

Theorem 11.1 (2-Parameter Depressed-Cubic Surface Family)**.** *In the general separated equation (25) of Theorem 7.2,*

$$2\left(y^2+3x^2\right)^2=\left[(a+p+e)^4-a^4-p^4-e^4\right]-\left[(2x+p)^4-(2x)^4-p^4\right],$$

setting $a=2x+q$ with

$$q=-\frac{e(e+2p)}{2(e+p)} \tag{36}$$

*eliminates both the x^4 and x^2 terms identically for **every** pair $(e,p)\in\mathbb{Q}^2$ $(e+p\neq 0)$, reducing the equation to the 2-parameter family of depressed-cubic surfaces:*

$$\left(y^2+3x^2\right)^2=16ex^3+\frac{e^2\left(e^2+4ep+6p^2\right)}{e+p}x+\frac{e^2\left(4p^4+4ep^3-e^2p^2-4e^3p-2e^4\right)}{4(e+p)^2}, \tag{37}$$

where (a,b,c,d,e) is reconstructed rationally via $a=2x-\frac{e(e+2p)}{2(e+p)}$, $b=-x+y$, $c=-x-y$, $d=2x+p$, and $e=e$.

Proof. Substituting $a=2x+q$ into the right-hand side of (25) and dividing by 2 yields a cubic polynomial in x with leading term $16ex^3$ and x^2 coefficient $12(e^2+2ep+2eq+2pq)=12\big[e(e+2p)+2(e+p)q\big]$. Setting $q=-\frac{e(e+2p)}{2(e+p)}$ makes the x^2 coefficient vanish identically and simplifies the x^1 and x^0 coefficients to the stated rational expressions. □

12 The Centered Double-Depressed-Cubic Identity (Global Canonical Form)

We now unify all previous reductions into a global canonical form for the threefold V.

Theorem 12.1 (Centered Double-Depressed-Cubic Canonical Form)**.** *For any (a,b,c,d,e) with sum $s=a+b+c+d+e$, define the centered coordinates*

$$u=a+b+c,\quad v=a-\frac{u}{2}=\frac{a-b-c}{2},\quad r=\frac{s+d}{2}=d+\frac{u+e}{2},\quad x=\frac{2v-u}{4}=-\frac{b+c}{2},\quad y=\frac{b-c}{2}, \tag{38}$$

with inverse linear reconstruction

$$a=v+\frac{u}{2},\quad b=-x+y,\quad c=-x-y,\quad d=r-\frac{u+e}{2},\quad e=e,\quad s=r+\frac{u+e}{2}.$$

*Then the Diophantine equation $a^4+b^4+c^4+d^4+e^4=s^4$ is **identically equivalent** to the Centered Double-Depressed-Cubic Identity:*

$$2\left(y^2+3x^2\right)^2=(u+e)r\left(4r^2+(u+e)^2\right)-uv\left(4v^2+u^2\right)-e^4. \tag{39}$$

Consequently:

1. *Setting $r=mv$ gives a 3-parameter family of depressed cubics in v (with zero v^2 term) for all $(u,e,m)\in\mathbb{Q}^3$:*

$$2\left(y^2+3x^2\right)^2=4\left[(u+e)m^3-u\right]v^3+\left[(u+e)^3m-u^3\right]v-e^4.$$

2. *Whenever $\frac{u}{u+e}=m^3$ (i.e., $u=\frac{m^3}{1-m^3}e$), both the v^3 and v^2 terms vanish simultaneously, leaving a purely linear right-hand side in v:*

$$2\left(y^2+3x^2\right)^2=\frac{m(1-m^8)}{(1-m^3)^3}e^3v-e^4.$$

Proof. Since $s=r+\frac{u+e}{2}$ and $d=r-\frac{u+e}{2}$, the even powers r^4 and r^2 cancel in s^4-d^4, leaving

$$s^4-d^4=\left(r+\frac{u+e}{2}\right)^4-\left(r-\frac{u+e}{2}\right)^4=(u+e)r\left(4r^2+(u+e)^2\right).$$

Likewise, since $a=v+\frac{u}{2}$ and $-(b+c)=2x=v-\frac{u}{2}$, we have $a^4-(b+c)^4=uv(4v^2+u^2)$, while $b^4+c^4=2(y^2+3x^2)^2-(b+c)^4$. Adding $a^4+b^4+c^4=2(y^2+3x^2)^2+uv(4v^2+u^2)$ to d^4+e^4 and equating to s^4 yields (39). □

13 The Paired-Solution Surface ($u=2e$) and Weierstrass Fibrations

Theorem 12.1 explains why several solutions in Tables I and II occur in **pairs** sharing identical values of e and $s-d=u+e$:

- **Pair 1** ($u=16$, $e=8$, $s-d=24$): Row 3 ($s=15$, $d=-9$) has $(r,v,x,y)=(3,-2,-5,9)$, and Row 4 ($s=135$, $d=111$) has $(r,v,x,y)=(123,18,5,-97)$.
- **Pair 2** ($u=208$, $e=104$, $s-d=312$): Row 16 ($s=435$, $d=123$) has $(r,v,x,y)=(279,90,-7,361)$, and Row 17 ($s=515$, $d=203$) has $(r,v,x,y)=(359,40,-32,426)$.
- **Pair 3 in Table II** ($u=695$, $e=-70$, $s-d=625$): Row 2 of Table II ($s=413$, $d=-212$) has $(r,v,x,y)=\left(\frac{201}{2},-\frac{195}{2},-\frac{445}{2},-\frac{335}{2}\right)$, and Row 3 of Table II ($s=507$, $d=-118$) has $(r,v,x,y)=\left(\frac{389}{2},-\frac{995}{2},-\frac{845}{2},-\frac{55}{2}\right)$.

Proposition 13.1 (The Parameter-Free Paired-Solution Surface $u=2e$)**.** *Both Pair 1 and Pair 2 satisfy $u=2e$ ($s-d=3e$). Substituting $u=2e$ into (39) and normalizing by e via $(R,V,Y)=\left(\frac{r}{e},\frac{v}{e},\frac{y}{e}\right)$ collapses all four solutions onto the single depressed-cubic surface*

$$2\left(Y^2+\frac{3}{4}(V-1)^2\right)^2=12R^3+27R-8V^3-8V-1, \tag{40}$$

at $(R,V,Y)=\left(\frac{3}{8},-\frac{1}{4},\frac{9}{8}\right)$ (Row 3), $\left(\frac{123}{8},\frac{9}{4},\frac{97}{8}\right)$ (Row 4), $\left(\frac{279}{104},\frac{45}{52},\frac{361}{104}\right)$ (Row 16), and $\left(\frac{359}{104},\frac{5}{13},\frac{213}{52}\right)$ (Row 17). Because $X=\frac{V-1}{2}$ is independent of R, fixing $V=V_0$ reduces (40) to a Weierstrass cubic $2W^2=12R^3+27R-(8V_0^3+8V_0+1)$ in (R,W) with $W=Y^2+\frac{3}{4}(V_0-1)^2$.

14 Explicit Pencils on the Four-Squares Surface S_0 ($2a+b+c=0$)

The Four-Squares Surface S_0 of Theorem 2.2,

$$\left(y^2+3x^2\right)^2+\left(w^2+3z^2\right)^2+x^2(x+4z)^2=\left(3x^2+2xz+4z^2\right)^2,$$

contains Row 1 at $(x,y,z,w)=(-3,\pm1,-1,\pm3)$, Row 21 at $(112,\pm424,-681,\pm557)$, and the boundary points $(0:0:1:\pm1)$.

Proposition 14.1 (Two Explicit Pencils of Genus-1 Curves on S_0)**.**

1. ***The Pencil*** $w=z+tx$ ***Through*** $(0:0:1:1)$*: Substituting $w=z+tx$ (so $d=tx$ and $e=-2z-tx$) into S_0 cancels the z^4 term identically and factors out x, leaving the cubic bundle in z:*
$$\left(y^2+3x^2\right)^2 = x\left[(8-t^4)x^3-4(t^3-1)x^2z-12(t^2-1)xz^2-16(t-1)z^3\right], \tag{41}$$
which passes through Row 1 at $t=-\frac{4}{3}$ and $t=\frac{2}{3}$ with $(x,y,z)=(-3,\pm1,-1)$.

2. ***The Even Biquadratic Pencil*** $w=ky$*: Because S_0 involves only y^2 and w^2, setting $w=ky$ produces a quadratic equation in $Y=y^2$:*
$$(k^4+1)Y^2+6\left(x^2+k^2z^2\right)Y+\left(x^4-4x^3z-12x^2z^2-16xz^3-7z^4\right)=0,$$
whose discriminant gives the genus-1 family $(k^4+1)y^2=-3(x^2+k^2z^2)+V$ with
$$V^2=(8-k^4)x^4+4(k^4+1)x^3z+6(2k^4+3k^2+2)x^2z^2+16(k^4+1)xz^3+(16k^4+7)z^4,$$
having rational points at $k=3$ (Row 1) and $k=\frac{557}{424}$ (Row 21).

15 The Opposite-Pair Surface ($e=-d$) and Its Sum-of-Fourth-Powers Identity

Theorem 15.1 (Opposite-Pair Surface Identities and Osculating Map)**.** *When $e=-d$ (so $d+e=0$ and $s=a+b+c=a-2x$, as in Row 1), the fourth-power equation $a^4+b^4+c^4+2d^4=(a+b+c)^4$ satisfies two equivalent identities:*
$$\left(y^2+3x^2\right)^2+d^4=4x(2x-a)\left(a^2-ax+2x^2\right), \tag{42}$$
$$\left(y^2+3x^2\right)^2+d^4+a^4=\left(4x^2-2ax+a^2\right)^2. \tag{43}$$
Expanding around Row 1 normalized at $\left(\frac{3}{4},-1,-1,-\frac{1}{2},\frac{1}{2}\right)$ via $a=\frac{3}{4}x+a_2z$, $b=-x+y$, $c=-x-y$, $d=-\frac{1}{2}x+d_2z$, $e=-d$, the 6-point osculating parabola $y^2=z(\beta x+\gamma z)$ of Theorem 6.1 restricts on $e_2=-d_2$ to:
$$\begin{aligned}
\beta(a_2,d_2) &= -\frac{19a_2-2d_2}{24}, \qquad \gamma(a_2,d_2) = -\frac{1121a_2^2+152a_2d_2+284d_2^2}{3456},\\
R_1(a_2,d_2) &= -\frac{139915a_2^3+1290a_2^2d_2+16644a_2d_2^2-84680d_2^3}{41472},\\
R_0(a_2,d_2) &= -\frac{3813697a_2^4+1033912a_2^3d_2+659736a_2^2d_2^2+161792a_2d_2^3+645904d_2^4}{11943936},
\end{aligned}$$
with $\frac{x}{z}=-\frac{R_0(a_2,d_2)}{R_1(a_2,d_2)}$ and $\left(\frac{y}{z}\right)^2=\frac{\gamma R_1-\beta R_0}{R_1}$.

Proof. Setting $z=0$ and $w=d$ in the Three-Squares Identity (4) immediately gives $(y^2+3x^2)^2+d^4-(4x^2-2ax+a^2)^2=-a^4$. Specializing Theorem 6.1 to $e_2=-d_2$ yields the stated homogeneous polynomials in (a_2,d_2). □

16 The Equal-Pair Surface ($d=e$) and Dual Osculating Families

Proposition 16.1 (Structure of the Equal-Pair Surface $d=e$)**.** *When $d=e$ (satisfied by Row 2: $s=15$, $(13,12,-6,-2,-2)$ and Row 5: $s=135$, $(111,114,-58,-16,-16)$):*

1. *In the Centered Double-Depressed-Cubic Identity (39), $d=e$ fixes $r=\frac{u+3e}{2}$, reducing the surface to*
$$2\left(y^2+3x^2\right)^2=(u+2e)^4-2e^4-uv\left(4v^2+u^2\right). \tag{44}$$
In the affine chart $d=e=-2$, Row 2 lies at $(a,x,y,u,v)=\left(13,-3,9,19,\frac{7}{2}\right)$ and Row 5 (divided by 8) lies at $\left(\frac{111}{8},-\frac{7}{2},\frac{43}{4},\frac{167}{8},\frac{27}{8}\right)$, with secant displacement satisfying $\Delta y=2\Delta a$.

2. *Placing the equal pair in (b,c) so that $y_0=0$ normalizes Row 2 to $\left(\frac{13}{2},-1,-1,6,-3\right)$ and Row 5 to $\left(\frac{111}{16},-1,-1,\frac{57}{8},-\frac{29}{8}\right)$, providing two additional base points for the 2-parameter osculating-parabola map of Theorem 6.1.*

17 The Hyperbolic $(s-d)(s-e)$ Factorization on the Three-Squares Identity

Theorem 17.1 (Hyperbolic $(s-d)(s-e)$ Factorization)**.** *In the Three-Squares Identity (4), imposing the relation*

$$\left[4(x^2+xz+z^2)-2a(x+z)+a^2\right]-\left(w^2+3z^2\right)=2ax+2m^2$$

cancels $2ax$ and a^2 identically to yield the hyperbolic equation

$$(z+2x-a)^2-w^2=(s-d)(s-e)=2m^2, \tag{45}$$

which is solved rationally for all (a,x,m,r) $(r\neq 0)$ by

$$z+2x-a=\frac{r^2+2m^2}{2r},\qquad w=\frac{r^2-2m^2}{2r}\qquad \left(s-d=r,\quad s-e=\frac{2m^2}{r}\right), \tag{46}$$

while the fourth-power equation reduces to the quadratic in z:

$$\left(y^2+3x^2\right)^2=(4mz+m(2x-a))^2+3m^2(2x-a)^2-4m^4-\frac{a^4+(2x-a)^4}{2}. \tag{47}$$

Proof. Since $4(x^2+xz+z^2)-2a(x+z)+a^2-(w^2+3z^2)=(z+2x-a)^2-w^2+2ax$, equating to $2ax+2m^2$ gives $(z+2x-a)^2-w^2=2m^2$. Because $d=-z+w$, $e=-z-w$, and $s=a-2x-2z$, we have $z+2x-a-w=-(s-e)$ and $z+2x-a+w=-(s-d)$. Substituting $w^2=(z+2x-a)^2-2m^2$ into (4) yields the quadratic in z. □

18 Explicit Polynomial Parametrization for the $e_0=\frac{1}{3}$ Sector (Row 7)

Theorem 18.1 (Explicit Polynomial Parametrization for the $e_0=\frac{1}{3}$ Sector)**.** *In Row 7 of Table I, $(a_0,b_0,c_0,d_0,e_0^*)=(-18,146,-121,182,16)$ with $s=205$ satisfies*

$$\frac{a_0+b_0+c_0}{a_0+b_0+c_0+d_0}=\frac{-18+146-121}{-18+146-121+182}=\frac{7}{189}=\frac{1}{27}=\left(\frac{1}{3}\right)^3\implies e_0=\frac{1}{3}.$$

Setting $z=189$, $x=\frac{1}{2}X$, and $y=\frac{1}{2}Y$ on $H_{1/3}$ and parameterizing the resulting conic $\left(3X^2+Y^2\right)^2=237216X^2+103398400X+7789670896$ around $(X_0,Y_0,U_0)=(-25,267,73164)$ via $3X^2+Y^2=73164+16u(X+25)$ yields the explicit polynomial parametrization:

$$\begin{aligned}
a(u)&=-432u^2-219492u+8981952,\\
b(u,W)&=\frac{3\left(200u^2+73164u-3045875\right)+3W}{2},\\
c(u,W)&=\frac{3\left(200u^2+73164u-3045875\right)-3W}{2},\\
d(u)&=546\left(8u^2-7413\right),\\
e(u)&=384u^2-73164u+2504726,
\end{aligned} \tag{48}$$

where (u,W) lies on the genus-1 curve

$$\mathcal{E}_{1/3}:\quad W^2=-4802496u^4+278353600u^3-12403862688u^2+997806276600u-23811513788559, \tag{49}$$

with base point at the tangent slope $u_0=\frac{58865}{2613}$.

Proof. This follows directly from Theorem 5.1 with $e_0=\frac{1}{3}$, $z_0=189$, $X_0=2x_0=-25$, $Y_0=2y_0=267$, and $t=16u$, noting that $237216=16^2\cdot 927$ cancels a factor of 32 in the rational reconstruction of $X(u)$ and $U(u)$. □

19 Secant-Interpolation Cubic and Slope Map on the Paired-Solution Surface ($u=2e$)

Theorem 19.1 (Secant Factorization and Rational Slope Parametrization on $u=2e$)**.** *In Proposition 13.1, substituting $V=2X+1$ (where $X=\frac{x}{e}=-\frac{b+c}{2e}$) brings the Paired-Solution Surface $u=2e$ into the form*

$$2\left(Y^2+3X^2\right)^2=12R^3+27R-\left(64X^3+96X^2+64X+17\right), \tag{50}$$

on which Row 3 and Row 4 have opposite X-coordinates $X=\mp\frac{5}{8}$:

$$\textit{Row 3:}\quad (R,X,Y)=\left(\frac{3}{8},-\frac{5}{8},\pm\frac{9}{8}\right),\qquad \textit{Row 4:}\quad (R,X,Y)=\left(\frac{123}{8},+\frac{5}{8},\pm\frac{97}{8}\right).$$

1. ***Secant Interpolation**: Along the linear interpolant $R(X)=\frac{63}{8}+12X$ and $W(X)=Y^2+3X^2=\frac{1205}{16}+\frac{583}{5}X$ connecting Row 3 and Row 4, the cubic equation (50) factors completely over $\mathbb{Q}$ as*

$$\frac{1}{800}(8X-5)(8X+5)(258400X+169211)=0.$$

2. ***Universal Slope Parametrization Around Row 3**: Because the right-hand side of (50) has degree 3 in (R,X) while the left-hand side is $2W^2$ (degree 2 in $W=Y^2+3X^2$), expanding around Row 3 via $X=-\frac{5}{8}+t$, $R=\frac{3}{8}+mt$, and $W=\frac{39}{16}+\frac{19(27m-16)}{156}t$ makes $t=0$ a double root for every slope $m\in\mathbb{Q}$, yielding the rational parameterization:*

$$\begin{aligned}
X(m)&=-\frac{91260m^3-98901m^2+311904m-287104}{48672(3m^3-16)},\\
R(m)&=\frac{3}{8}+\frac{m\left(98901m^2-311904m-199616\right)}{48672(3m^3-16)},\\
789654528(3m^3-16)^2Y(m)^2&=24824357064m^6-41251258152m^5-69076393785m^4\\
&\quad-136173582528m^3+162207048960m^2\\
&\quad+191717867520m+309338767360.
\end{aligned}$$

Proof. Substituting $X=-\frac{5}{8}+t$ and $R=\frac{3}{8}+mt$ into the right-hand side of (50) divided by 2 gives $W_0^2+c_1(m)t+c_2(m)t^2+c_3(m)t^3$ with $W_0=\frac{39}{16}$, $c_1(m)=\frac{741(27m-16)}{128}$, $c_2(m)=\frac{9(9m^2-32)}{16}$, and $c_3(m)=6m^3-32$. Setting $W=W_0+\alpha(m)t$ with $\alpha(m)=\frac{c_1(m)}{2W_0}=\frac{19(27m-16)}{156}$ cancels the t^0 and t^1 terms of $W^2-\mathrm{RHS}=0$, leaving $t_{\mathrm{new}}(m)=\frac{\alpha(m)^2-c_2(m)}{c_3(m)}$, which produces the stated rational functions. □

20 Master Classification of Parametric Families and Table Solutions

The table below synthesizes how the algebraic surfaces and parametric constructions developed in Sections 1–16 classify the primitive solutions in Tables I and II.

Algebraic Surface / Family	**Defining Relation on** (a,b,c,d,e,s)	**Reduced Canonical Equation**	**Solutions in Census Belonging to Family**
Bitangent Hyperplane H_{-1} **(**$e_0=-1$**)**	$2(a+b+c)+d=0$	$(3x^2+y^2)^2 = z\big[24wx^2-12w(w+z)x+(2w^3+3w^2z+2wz^2-8z^3)\big]$	**Row 1** ($s=5$, $w=9z$) **Row 18** ($s=515$, $z=13^2, w=3\cdot 17^2$)
Bitangent Hyperplane H_2 **(**$e_0=2$**)**	$7(a+b+c)+8d=0$	$(3x^2+y^2)^2 = z\big[48(2w-15z)x^2+8(3w^2+3wz-255z^2)x+\dots\big]$	**Row 6** ($s=175$, $z=2, w=81$)
Bitangent Hyperplane $H_{1/3}$ **(**$e_0=\frac{1}{3}$**)**	$26(a+b+c)-d=0$	$(3X^2+Y^2)^2 = 237216X^2+103398400X+7789670896$	**Row 7** ($s=205$, $X=-25, Y=267$)
Bitangent Hyperplane $H_{3/5}$ **(**$e_0=\frac{3}{5}$, $5\nmid s$**)**	$98(a+b+c)-27d=0$	$(3X^2+Y^2)^2 = 147744X^2+21628032X+221539312$	**Row 4 of Table II** ($s=551$, $X=9, Y=143$)
Four-Squares Surface S_0	$2a+b+c=0$ $(a=x)$	$(y^2+3x^2)^2+(w^2+3z^2)^2+x^2(x+4z)^2=(3x^2+2xz+4z^2)^2$	**Row 1** ($s=5$, $(-3,1,-1,3)$) **Row 21** ($s=625$, $(112,424,-681,557)$)
Paired-Solution Surface $(u=2e)$	$s-d=3e$ $(a+b+c=2e)$	$2(Y^2+3X^2)^2 = 12R^3+27R-(64X^3+96X^2+64X+17)$	**Row 3 & Row 4** ($s=15,135$, $e=8$) **Row 16 & Row 17** ($s=435,515$, $e=104$)
Non-Divisible Paired Slice $(5\nmid s)$	$s-d=625$, $e=-70$	$2(y^2+3x^2)^2=625r(4r^2+625^2)-695v(4v^2+695^2)-70^4$	**Row 2 & Row 3 of Table II** ($s=413,507$)
Opposite-Pair Surface	$d+e=0$ $(e=-d)$	$(y^2+3x^2)^2+d^4+a^4=(4x^2-2ax+a^2)^2$	**Row 1** ($s=5$: $48^2+2^4+3^4=49^2$)
Equal-Pair Surface	$d=e$ $(w=0)$	$2(y^2+3x^2)^2 = (u+2e)^4-2e^4-uv(4v^2+u^2)$	**Row 2** ($s=15$, $d=e=-2$) **Row 5** ($s=135$, $d=e=-16$)

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